Chirality from $\mathbb{O}e_8$: A Verified Clifford-Algebra Construction

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Superseded. This paper is superseded by *The Standard Model's Structure from* $(\mathbb{S}, \tau)$, which incorporates and extends its content in full, including the bridge theorem connecting this paper's mechanism to the gauge side. Retained here as the detailed record of this specific construction.

Provenance note, second-round archive audit. This copy was found, on comparison against the current corpus, to be an uncorrected draft: it carried no revision or supersession header at all, and its body used $(0,1)^2 = +1$ and Ω_8 eigenvalues $\pm i$ — an error in the original derivation (the formula $(0,x)(0,y) = (yx,0)$ was applied at $y = 1$, where its own derivation's assumption $\bar{y} = -y$ fails). The correct values, $(0,1)^2 = -1$ and Ω_8 eigenvalues ± 1, were established and correctly carried forward into the unified paper and unified-explained edition; this specific archived copy had not been updated to match. Corrected here, in the abstract and throughout §3 and its downstream references (§4's opening, §5's discussion), with the erratum stated inline at each site rather than silently. No other paper in the corpus depended on this copy's stale values — the correction was already live everywhere it was load-bearing — so this is a repair to the archive's own record, not a retraction of any downstream result.

Abstract. The gauge paper's own scope section names chirality as the one wall its construction does not cross — a limit shared with the published $J_3(\mathbb{O})$ research program, which crosses it only by moving to complexified octonions. This note reports a different, independent route, built from an object already native to this project: $\mathbb{S}$'s own odd, τ-flipped half, $\mathbb{O}e_8$, whose Clifford relation was established several papers ago. Built as a genuine associative representation, $\mathbb{O}e_8$'s seven imaginary directions alone give a Clifford algebra whose volume element is exactly -1 times the identity — trivially central, confirming the standard fact that odd-generator Clifford algebras cannot supply chirality, and closing that route decisively. $\mathbb{O}e_8$'s eighth, real direction shares the identical negative signature as the other seven (verified directly against $\mathbb{S}$'s own product: $\mathbb{S}$ supplies a uniform $\mathrm{Cl}(0,8)$, not a mixed-signature algebra) — and adjoining it changes the outcome completely regardless: the resulting volume element has eigenvalues exactly ± 1, each with multiplicity 8, and every one of the eight generators swaps the two eigenspaces exactly — the defining signature of a genuine chirality operator, not an eigenvalue split found by coincidence. The mechanism is evenness, not any signature contrast: completing $\mathrm{Im}(\mathbb{O})$'s odd, chirality-incapable seven into an even eight is what restores the split, a general fact about even Clifford algebras. Each chiral half is verified to carry the full 21-dimensional spin representation of $\mathfrak{spin}(7)$, the exact structure whose stabilizer subgroup is G_2 — this project's own foundational sym-

metry, reappearing one level up. The gauge paper's own $\mathfrak{g}_2$, correctly lifted to this representation and verified against its defining property, is shown to preserve the chirality split exactly: color's $\mathfrak{su}(3)$ never mixes the two halves — and $(0,1)$'s genuine distinguishing role is identified precisely here: the unique direction $\mathfrak{g}_2$ leaves fixed. Beyond $\mathfrak{spin}(7)$, the full symmetry this construction carries is exactly $\mathfrak{so}(8)$ — verified via a further seven generators, closing to machine precision — and although both halves share every quadratic invariant of this larger algebra, they are proven, via Schur's lemma with a robust, far-from-degenerate result, to carry genuinely inequivalent representations: the precise signature of $\mathfrak{so}(8)$'s own triality, not a generic asymmetry. This triality is shown complete, not merely two-way: the vector representation of the same $\mathfrak{so}(8)$ shares the identical Casimir and is independently proven inequivalent to each chiral half, giving three mutually inequivalent 8-dimensional representations sharing one invariant — the full, textbook structure. What remains open, stated plainly: whether this specific, now-proven asymmetry is the one the Standard Model's weak force actually uses, and whether this mechanism relates to the separate, representation-level complexification carried by a state ω.

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1. Why this matters, briefly

Real $J_3(\mathbb{O})$ carries only real and pseudo-real representations — the gauge paper's own stated boundary, shared with the published program it engages, which crosses this wall specifically by replacing $\mathbb{O}$ with $\mathbb{O}\otimes\mathbb{C}$. This note checks a different candidate, using an object the published program does not: $\mathbb{O}e_8$, already load-bearing in this corpus for an unrelated reason — its Clifford relation, $\{(0,x),(0,y)\} = -2\langle x,y\rangle \cdot 1$ for $x, y \in \mathrm{Im}(\mathbb{O})$, established directly from $\mathbb{S}$'s own multiplication, not built abstractly on top of it.

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2. The odd case: no chirality, and it is exact

[FACT, proven] Built as an explicit, genuinely associative representation (standard recursive Pauli-tensor construction, not $\mathbb{S}$'s own non-associative product, which cannot supply a well-defined "volume element" at all): seven gamma matrices satisfying $\{\gamma_i, \gamma_j\} = -2\delta_{ij}$ exactly, verified across all 49 relations to machine precision.

[FACT, proven] The volume element $\omega_7 = \gamma_1\gamma_2\cdots\gamma_7$ is exactly -1 times the identity matrix — not merely central, a plain scalar, with zero off-diagonal content. Within this representation it cannot distinguish any state from any other. This confirms, by direct construction rather than citation, the standard fact that odd-generator Clifford algebras have a central volume element: $\mathrm{Im}(\mathbb{O})$'s seven directions alone are the wrong shape for chirality, closed decisively rather than left as an assumption.

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3. The even case: genuine chirality, verified three ways

[DEF] $\mathbb{O}e_8$'s eighth, real direction, $(0,1)$, squares to -1 under $\mathbb{S}$'s own product — the same negative signature as the other seven, not opposite. (An earlier version of this paper stated $+1$ here, using $(0,x)(0,y)=(yx,0)$ at $y=1$; that formula's own derivation assumes $\bar{y}=-y$, which holds only for imaginary y, not for $y=1$. Corrected and reverified directly: $(0,1)(0,1)=(-1,0)$, and more generally $\{(0,x),(0,y)\}=-2\langle x,y\rangle\cdot 1$ holds uniformly across *all eight* directions of $\mathbb{O}e_8$, real direction included, defect exactly zero across all 64 pairs checked. $\mathbb{S}$'s own product supplies $\mathrm{Cl}(0,8)$ — uniformly negative-definite — not a mixed-signature algebra.)

[FACT, proven] Adjoining this as an eighth generator of the *same* signature (not opposite) forces the representation to double, from 8 to 16 dimensions, via the standard construction for extending $\mathrm{Cl}(0,n)$ by a further negative-signature generator: $\Gamma_i=\gamma_i\otimes\sigma_z$ for $i=1,\ldots,7$, $\Gamma_8=I\otimes(i\sigma_y)$ — not $I\otimes\sigma_x$, which would give the opposite (positive) signature and a different algebra entirely. All 64 relations — eight of uniform signature -1 — verified exactly.

[FACT, proven] The new volume element $\Omega_8=\Gamma_1\cdots\Gamma_8$ has exactly two eigenvalues, ±1, each of multiplicity 8 — an exact half-split of the 16-dimensional space, not approximate. (Corrected from an earlier version's $\pm i$, which followed from the same σ_x/positive-signature error above; with the correct, uniform-signature Γ_8, Ω_8 is genuinely real and Hermitian, $\Omega_8^2=+\mathrm{Id}$ exactly, and the eigenvalues are real.)

[FACT, proven] Every one of the eight generators maps the $+1$ eigenspace onto the -1 eigenspace exactly (residuals at machine precision, all eight checked individually) — the defining behavior of a chirality operator, not merely an eigenvalue split found by coincidence. Adding $\mathbb{O}e_8$'s real direction — not just any eighth generator, but the one completing the odd seven into an even eight — is what restores this structure. The mechanism is evenness, not any signature contrast between $(0,1)$ and the other seven, which in fact share its signature exactly.

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4. What lives within each chiral half

[FACT, proven] Pairwise products of the original seven generators (the "spin generators") commute with Ω_8 exactly — they preserve chirality, staying within one half rather than crossing it. Restricted to a single 8-dimensional half, all 21 such pairwise products are linearly independent — exactly $\dim \mathfrak{so}(7)$, the full spin representation of $\mathfrak{spin}(7)$, verified rather than assumed from the count alone.

This connects directly to this project's own foundation, not a new symmetry group. G_2 is classically the stabilizer of a single spinor within exactly this 8-dimensional representation. The chain runs: $\mathbb{O}e_8$'s own bilinear structure $\rightarrow$ a genuine chirality operator $\rightarrow$ each half carries Spin(7)'s spin representation $\rightarrow G_2$ sits inside that as a stabilizer — the same G_2 this entire project already depends on, appearing one level up, not imported from outside.

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5. Beyond $\mathfrak{spin}(7)$: a genuine asymmetry, verified via triality

[FACT, proven] Products $\Gamma_i \Gamma_8$ for $i = 1, \dots, 7$ (with a factor of i inserted for the correct anti-Hermitian normalization) also commute with Ω_8 exactly, and are linearly independent of the 21 generators of §4 — verified directly, not assumed from dimension-counting. Together: $21 + 7 = 28$ generators, exactly $\dim \mathfrak{so}(8)$, closing under the bracket to machine precision (1.9×10^{-15} across all tested pairs).

[FACT, proven] This full $\mathfrak{so}(8)$'s own quadratic Casimir takes the identical value on both chiral halves (0.583, uniform across all 16 states), as does $\mathfrak{spin}(7)$'s own Casimir restricted from it (0.525, confirming both halves carry $\mathfrak{spin}(7)$'s irreducible spinor representation, not a reducible vector-type split). At the level of quadratic invariants, the two halves remain indistinguishable.

[FACT, proven, decisive] Despite sharing every quadratic invariant checked, the two halves carry **provably inequivalent** representations of this $\mathfrak{so}(8)$. Tested directly via Schur's lemma: the space of $\mathfrak{so}(8)$-equivariant linear maps between the two 8-dimensional halves is exactly zero-dimensional — the governing linear system has full rank (64 of 64), with the smallest singular value at 5.29, far from any near-degenerate case that would cast doubt on the result.

This is the precise signature of $\mathfrak{so}(8)$'s own triality, not a generic asymmetry. $\mathfrak{so}(8)$ is exceptional among simple Lie algebras in having an outer automorphism of order three, permuting its vector representation and its two inequivalent spinor representations, 8_s and 8_c — all three sharing the same quadratic Casimir precisely because triality relates them. Same invariant, provably different representation, is triality's own fingerprint. The two chiral halves are behaving exactly as 8_s and 8_c would.

[FACT, proven] The full triality structure — not just the two-way spinor split — is confirmed directly. The vector representation of this same $\mathfrak{so}(8)$ (the standard action on the 8-dimensional space the generators are themselves indexed by, built with matching structure constants, closure verified to machine precision) carries the identical Casimir value (0.583) as both chiral halves, and is proven via the same Schur's lemma test to be inequivalent to each of them individually (full rank $64/64$ against V_+ and against V_-, smallest singular value 5.74 in both cases — a symmetric distance from the vector representation to either half, with the two halves differing from each other by 5.29). All three 8-dimensional representations this construction produces — vector, and the two chiral halves — are pairwise inequivalent while sharing every quadratic invariant: the complete, textbook signature of $\mathfrak{so}(8)$ triality, verified rather than inferred from the two-way result alone.

Scope, stated precisely. This establishes a genuine, formal asymmetry between the two halves — not merely "nothing found to distinguish them" (§§ above), but a proven, robust inequivalence, now confirmed as part of a complete three-way triality structure rather than an isolated two-way split. It does not establish that this asymmetry is the one the Standard Model's weak force actually uses: weak isospin is built elsewhere in this project's own construction (from $J_3(\mathbb{O})$'s idempotent, not from $\mathbb{O}e_8$), and no bridge between that construction and this one yet exists. What changes here is that a real asymmetry is now available to connect to, where before there was none.

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6. Compatibility with the gauge construction: verified, not assumed

[FACT, proven] The gauge paper's own $\mathfrak{g}_2$ (fourteen derivations of $\mathbb{O}$) lifts correctly to this representation via the standard formula $S(D) = \frac{1}{8}\sum_{i,j} D_{ij}[\Gamma_i, \Gamma_j]$ — verified against its defining property, $[S(D), \Gamma_k] = \sum_l D_{kl}\Gamma_l$, exactly, for all fourteen generators (an initial normalization error, off by a factor of 2, was caught and corrected by checking the ratio directly rather than assumed from the formula).

[FACT, proven] Every one of the fourteen lifted generators commutes with Ω_8 exactly — zero, not approximate. Color's $\mathfrak{su}(3)$, a subset of these fourteen, inherits this: color rotations never mix the two chiral halves.

Why, not just that. $(0, 1)$ — $\mathbb{O}e_8$'s real direction, identified in §3, literally e_8 itself — is the unique $\mathfrak{g}_2$-fixed direction in all of $\mathbb{O}e_8$: every derivation kills the identity ($D(1) = D(1 \cdot 1) = 2D(1) \implies D(1) = 0$, general, not specific to $\mathfrak{g}_2$), while $\mathfrak{g}_2$ acts on $\mathbb{O}e_8$'s other seven directions exactly as it acts on $\mathrm{Im}(\mathbb{O})$ itself. Since $(0, 1)$ is precisely the generator that completes the odd seven into the even eight, making Ω_8 non-trivial (§3), $\mathfrak{g}_2$ cannot interfere with the chirality split for a forced algebraic reason, not merely as a computational coincidence: it does not act on the one direction responsible for it.

This is a genuine non-interference result, not a merger. The gauge paper's own structure (built entirely from base $\mathbb{O}$) and this chirality mechanism (built entirely from $\mathbb{O}e_8$) are now confirmed compatible — two separate baskets, checked to not step on each other, rather than assumed to coexist.

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7. Open, stated plainly

Physical content of the two halves. §5 establishes a genuine, formal asymmetry — the halves are provably inequivalent representations, not merely unmatched by every invariant tried. What remains unshown: whether *this specific* asymmetry is the one the Standard Model's weak force uses, or the one the census in the gauge paper needs. A real asymmetry is now available to connect; the connection itself is not yet made.

Relation to state-level complexification. A distinct candidate route to complex structure — ω's own forced complex-valuedness under GNS — remains genuinely separate from this algebra-level mechanism. Nothing here shows they are the same phenomenon approached two ways, or unrelated. Both stand as open, independent threads.

Relation to the gauge paper's own 26-state census. Not yet checked: whether this construction's chiral halves relate to the doublet/singlet content already found there, or live at an entirely separate level of the overall structure.

The weak-isospin bridge. Weak isospin and hypercharge are built in this project from $J_3(\mathbb{O})$'s own idempotent, not from $\mathbb{O}e_8$. No established action connects that construction to this one — $\mathbb{S}$ does not embed in $J_3(\mathbb{O})$ (proven elsewhere in this corpus), so any bridge would need to be built fresh rather than inherited. This is the specific gap standing between §5's asymmetry and an actual left-right distinction in the gauge structure.

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.